



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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SECJ1023 - PROGRAMMING TECHNIQUE II

SECTION 02

PROJECT 1

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Introduction:

The objective of this proposal is to suggest the development of an OOP-based hotel reservation system that employs the use of abstraction, polymorphism, inheritance, encapsulation, and association. This system will be designed to assist customers and administrators manage hotel reservations more effectively and efficiently.

The Hotel Reservation System is a software solution that simplifies the booking process for both customers and administrators, increasing efficiency and improving the overall guest experience. The system allows the user to log in as either a customer or an administrator. The customer menu allows customers to enter their personal information, which is then processed through multiple functions to display available rooms, calculate total booking prices, and ultimately view their booking receipts. The admin menu, on the other hand, allows administrators to log into the system and manage all bookings, ensuring accurate records and efficient room allocation. Furthermore, the administrator can update room availability and handle cancellations or maintenance requests. Additionally, the administrator can set room prices based on demand, monitor revenue, and adjust pricing strategies. The system's user interface will be intuitive and easy to use. Overall, the Hotel Reservation System, which will be built in C++, will offer a comprehensive solution for both customers and administrators to efficiently manage reservations and administrative tasks.

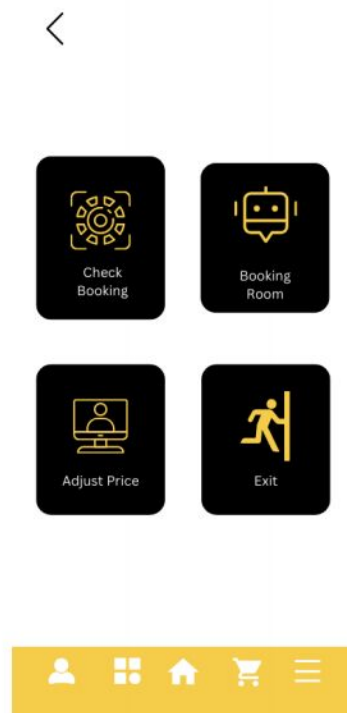
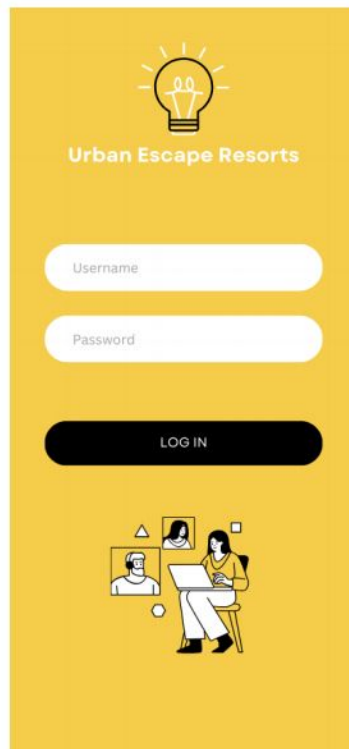
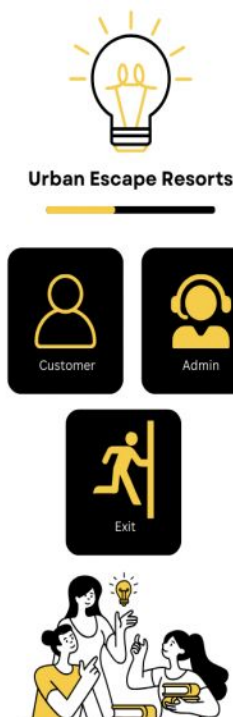
Solution:

The system that is currently under development will be written in C++ and include a number of functions for handling hotel reservations based on the supplied information. The system will include such things as user, admin, room booking, the availability of rooms, and a price list for each room. The hotel reservation system will be easily managed with the cooperation of these courses. The user will have the ability to view the pricing list, room type, and availability for all rooms, but first they must register their personal information. Additionally, users can also select the number of rooms and days they would like to reserve. Once everything is finalized, they can select their preferred method of payment and leave feedback in order for the hotel to enhance their stay once they have been accommodated. The same procedures will be followed even if the admins desire to reserve the room. However, admins have the ability to manage the rooms reservation and the price of each type of room. At the same time, admins are granted access to both daily and monthly views of the overall revenue.

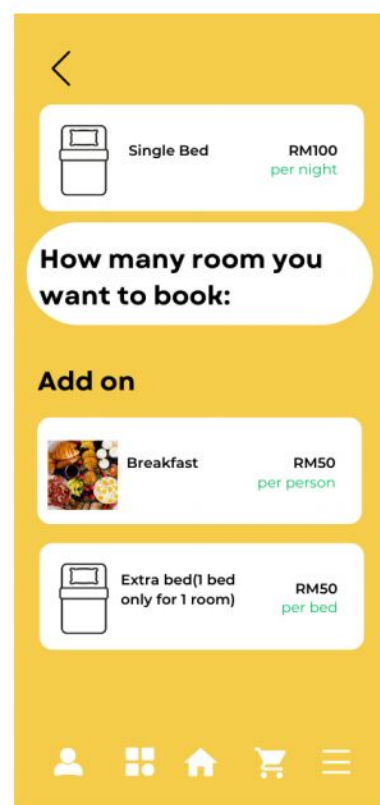
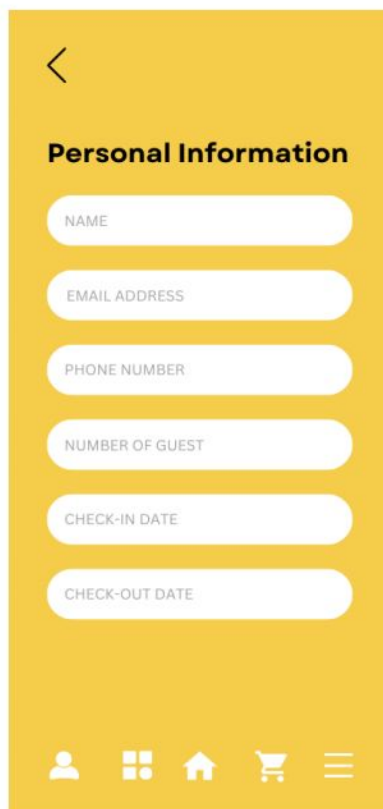
Conclusion:

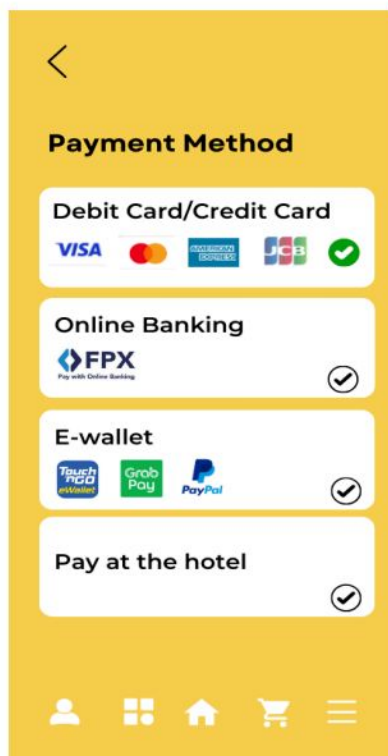
In conclusion, the C++ foundation of our hotel reservation system allows users to make reservations smoothly, while admins could manage the reservation, availability of the room, and price list, as well as prevent someone from booking a room but unintentionally canceling it by the system. With its well-developed and user-friendly features, by applying OOP principles, hotel reservations will be more efficient, secure, and straightforward..

Storyboard-Admin

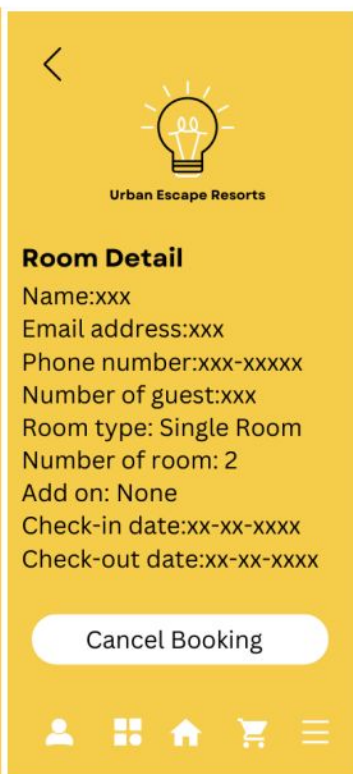
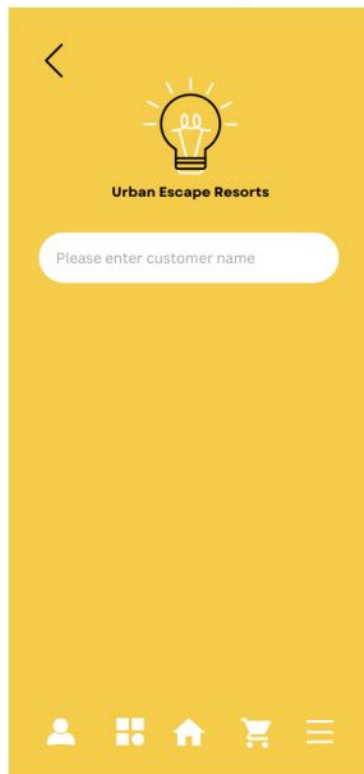


Booking Room





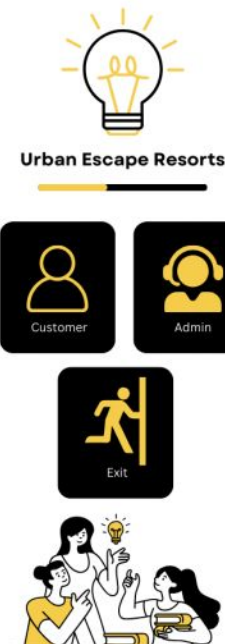
Check booking room



Adjust Price



Storyboard-Customer



Personal Information

NAME

EMAIL ADDRESS

PHONE NUMBER

NUMBER OF GUEST

CHECK-IN DATE

CHECK-OUT DATE

Room Type

Single Bed RM120 per night

Single Bed(with balcony) RM180 per night

Double Bed RM240 per night

Double bed(with balcony) RM300 per night

How many room you want to book:

Add on

Breakfast RM50 per person

Extra bed(1 bed only for 1 room) RM50 per bed

Payment Method

Debit Card/Credit Card

VISA Mastercard American Express JCB

Online Banking

FPX

E-wallet

Touch n Go Grab Pay PayPal

Pay at the hotel

Receipt

Name:xxx
Email Address:xxx
Phone Number:xxx
Number of guests:xxx
Check-in date:xxx
Check-out date:xxx
Room type:xx
Number of room:xxx

Payment information

Payment method: debit card
1 room x 2 night RM 400
Add on: RM 0
Total amount: RM 400

Thank you for choosing our hotel. 😊



Urban Escape Resorts

Review



Review and rate

Share details of your
experience at this hotel



